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Systematic Literature Review

Methods to Include Environmental Impacts in Health Economic Evaluations and Health Technology Assessments: A Scoping Review

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ABSTRACT

Objectives: The environmental impacts of healthcare are important factors that should be considered during health technology assessments. This study aims to summarize the evidence that exists about methods to include environmental impacts in health economic evaluations and health technology assessments.

Methods: We identified records for screening using an existing scoping review and a systematic search of academic databases and gray literature up to September 2023. We screened the identified records for eligibility and extracted data using a narrative synthesis approach. The review was conducted following the JBI Manual for Evidence Synthesis and reported according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses Extension for Scoping Reviews checklist.

Results: We identified 2898 records and assessed the full text of 114, of which 54 were included in this review. Ten methods were identified to include environmental impacts in health economic evaluations and health technology assessments. Methods included converting environmental impacts to dollars or disability-adjusted life years and including them in a cost-effectiveness, cost-utility, or cost-benefit analysis, calculating an incremental carbon footprint effectiveness ratio or incremental carbon footprint cost ratio, incorporating impacts as one criteria of a multi-criteria decision analysis, and freely considering impacts during health technology assessment deliberation processes.

Conclusions: Methods to include environmental impacts in health economic evaluations and health technology assessments exist but have not been tested for widespread use by health technology assessment agencies. Further research and implementation work is needed to determine which method can best aid decision makers to choose low environmental impact healthcare interventions.

Keywords: economic evaluation, environmental impact, healthcare sustainability, health technology assessment, review.

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Highlights

- Healthcare is a large source of greenhouse gas emissions and other environmental impacts, and these need to be considered during health technology assessments so that decision makers can prioritize low environmental impact options to reduce the healthcare sector's impact on climate change.
- We identified 10 methods to include environmental impacts in health technology assessments or the economic evaluations they rely on. These include converting environmental impacts to dollars or disability-adjusted life years and including them in a cost-effectiveness, cost-utility, or cost-benefit analysis, calculating an incremental carbon footprint effectiveness ratio or incremental carbon footprint cost ratio, incorporating impacts as one criteria of a multi-criteria decision analysis, and freely considering impacts during health technology assessment deliberation processes.
- There are established methods to identify, measure, and value the environmental impacts of healthcare technologies and interventions. Further research and implementation work is needed to determine which method can best aid decision makers to choose low-carbon healthcare.

Introduction

Climate change is the biggest health threat facing humanity.[1] Rising temperatures are increasing the frequency of extreme weather, shifting the geographic range of climate-sensitive diseases, affecting food and water security, worsening air quality, and disrupting the social systems on which health depends.[2] Limiting temperature increases requires rapid carbon dioxide (CO_2) emissions reductions in all sectors[3] and the healthcare sector is no exception to this.

Healthcare is responsible for 5% of global CO_2 emissions and is only surpassed by sectors related to energy, transport, and construction.[4,5] Most of these emissions come from the delivery of care itself.[6] In order to reduce healthcare's impact on climate change, clinical practices need to move to low-carbon options. At the United Nations Climate Change Conference in 2021, 50 countries committed to net-zero, low-carbon, or sustainable healthcare systems,[7] and there are many advocacy groups across the world working to achieve the same.[8-10]

The Centre for Sustainable Healthcare has proposed 4 principles of sustainable clinical practice to reduce the environmental impact of healthcare and improve health outcomes. These are disease prevention, patient empowerment, lean service delivery, and the preferential use of low-carbon alternatives.[11] To achieve this fourth principle, environmental impacts must be considered in purchasing and funding decisions. In this way, funders may choose care that is both cost-effective and has a low environmental impact.

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In many jurisdictions funding decisions are aided by a process known as Health Technology Assessment (HTA), which aims to determine the value of medicines, tests, and treatments.[12] HTA decisions typically rely on cost-effectiveness and budget impact studies that focus on the health benefits and health system costs of an intervention. Although not all healthcare jurisdictions have established HTA processes, they exist in many high- and upper-middle-income countries—jurisdictions responsible for most greenhouse gas emissions.[12]

HTA guidelines in Australia[13] and Canada[14] currently allow environmental impacts to be considered during decision making, and the National Institute for Health and Care Excellence in the UK have pledged to explore ways to incorporate environmental impact data in their guidance.[15] However, methods for including environmental impacts in HTA, or the economic evaluations they rely on, are underdeveloped, and there is limited guidance on the methods available and which one(s) should be used. The objective of this scoping review was to summarize current evidence on methods to include environmental impacts in health economic evaluations and HTAs.

Methods

This scoping review was conducted in accordance with the JBI Manual for Evidence Synthesis[16] and is reported according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses Extension for Scoping Reviews (Appendix A in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.02.019).[17] A detailed description of the review's objectives and methods were pre-specified in the review protocol.[18]

Review Question

What evidence exists on methods to measure and include environmental impacts in health economic evaluations and HTAs?

Inclusion Criteria

Records that met the following participant, concept, and context criteria were included in this scoping review:

- Participants: all participant types and health conditions, including research in which a health intervention or technology was investigated,
- Concept: methods, tools, guidance, examples, commentary, and all other records related to the inclusion of environmental impacts in health economic evaluations and HTAs, including research that presented environmental impacts alongside economic impacts,
- Context: all contextual settings (eg, countries, health systems, and settings).

We considered all quantitative study designs including economic evaluations alongside randomized controlled trials, economic models, and retrospective analyses. We also considered qualitative studies, opinion papers, and gray literature. Existing reviews and records that investigated the environmental impact of a technology but did not do so alongside economic impacts or in the context of an economic evaluation or HTA were not included.

Search Strategy

An initial limited search of MEDLINE and Scopus was undertaken to identify records relevant to the review question, and the keywords and index terms for these records were used to develop a full search strategy in consultation with an academic librarian. We identified an existing scoping review on this topic and used this as a starting point for our search.[19,20] The authors of the existing review searched PubMed, Embase, and Scopus from their inception to 2021. We believed this search to be comprehensive and considered all records included in their review for inclusion in the current review. In addition, we searched MEDLINE and Scopus for records published after the existing review, between 2021 and 6 Sep 2023. We also searched EconLit, Environment Complete, and Evidence-Based Medicine Reviews, which were not searched in the existing review, from their inception to 6 Sep 2023. The search strategies for each database are detailed in Appendix B in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.02.019. Gray literature was identified using the Grey Matters tool[21] and Google (https://www.google.com.au). The reference lists and forward citations of included records were hand searched using Scopus and Google Scholar (https://scholar.google.com). Records in any language were eligible for inclusion and translated using DeepL Translate (https://www.deepl.com).

Record Screening and Selection

All identified records were collated using EndNote X9 (Clarivate Analytics) and uploaded to Covidence (Veritas Health Innovation) for screening. Duplicate records were automatically removed. Title and abstract examinations were completed by 1 reviewer (J.T.W.W.). The full text of potentially relevant records were retrieved and independently examined by 2 reviewers (J.T.W.W. and M.D.). Reasons for exclusion after full text review were recorded. Disagreements were resolved by discussion, with the involvement of 2 additional reviewers where necessary (K.J.L.B. and R.L.M.).

Data Extraction

Relevant data were extracted from included records by 1 reviewer (J.T.W.W.) and checked by another reviewer (M.D.), using a data extraction tool developed by all authors. The extraction tool was refined after a pilot of 10 records and included authorship and publication details, objectives, a description of any relevant methods undertaken or discussed (including details about the technology being evaluated, the environmental impacts considered, and any conversions used), and details of the stated benefits and limitations of each method.

Analysis and Presentation of Results

We conducted a narrative synthesis of the included records to summarize findings relevant to the review question. Information about methods to identify, measure, and value environmental impacts were summarized first. Then, information about methods to include environmental impacts in an economic evaluation or HTA were categorized using the approaches proposed by Toolan et al.[22] and summarized in tables. Toolan et al.[22] propose 4 approaches to include environmental impacts in health technology assessments: parallel evaluation, integrated evaluation, information conduit, and environment-focused evaluation.[22] Parallel evaluations seek to include environmental impacts in value judgments alongside established health economic analyses without changes to existing economic methods.[22] Integrated evaluations seek to incorporate environmental impacts into the economic evaluation or HTA by synthesizing clinical, financial, and environmental information into 1 quantitative analysis.[22] An information conduit approach involves HTA agencies reporting environmental impacts alongside HTA outcomes without any attempt to integrate them into the decision or analysis.[22] Environment-focused evaluations apply to technologies that are not expected to have any additional health or cost benefits. HTA

agencies following this approach would only consider environmental impacts in their deliberations.[22] A similar method exists in the case of cost-minimization analysis, when the health benefits of technologies are deemed equivalent, and a decision is made on the basis of costs alone. We did not appraise the quality of included records because this is not required for scoping reviews.[23] A glossary of environmental terms is included in Box 1.

Box 1.
Glossary of environmental terms used in this review.

Term	Definition
Carbon dioxide (CO_2)	Carbon dioxide is the principal greenhouse gas contributing to climate change.
Carbon dioxide equivalent (CO_2-e)	Carbon dioxide equivalent is a metric used to summarize and/or compare different greenhouse gases or environmental impacts.
Greenhouse gases (GHG)	Greenhouse gases are those that contribute to the greenhouse effect and raise the Earth's surface temperature, including carbon dioxide, methane, ozone, and nitrous oxide.
Life cycle analysis (LCA)	Life cycle analysis is a method for assessing the environmental impacts associated with the entire life cycle of a product or process.
Process-based LCA	Process-based LCA is a "bottom up" approach to LCA that uses measures of the actual process being assessed to calculated environmental impact.
Social cost of carbon (SCC)	An estimate in monetary units of the damage caused by carbon dioxide to society.

Results

Search Results

We identified 2835 unique records from our database search and 63 from other sources, including the existing review. After title, abstract, and full text screening we included 54 records in this review (Fig. 1; a full list is available in Appendix C in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.02.019). Records were published between 2005 and 2024 and the majority since 2020 (n = 5 in 2020, n = 6 in 2021, n = 11 in 2022, and n = 9 in 2023, n = 1 in 2024). The remainder of the results section is organized by distinguishing records that identify, measure, and value environmental impacts in a health economic context and records that discuss or apply methods to include those impacts in an economic evaluation or HTA. These 2 groups are not mutually exclusive. Appendix C in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.02.019 contains a separate table for each type of record.

Methods to Identify, Measure, and Value the Environmental Impacts of Healthcare

Forty-eight records identified, measured, or valued environmental impacts in a health economic context.[24-71] These records frequently identified the greenhouse gas emissions, energy use, water use, and physical waste impacts of the intervention being assessed. Other environmental impacts identified included human and environmental toxicity, particulate matter, eutrophication, acidification, pharmaceutical and chemical contamination, land use, ozone depletion and formation, ionizing radiation, and the depletion of natural resources.

Measurements were frequently in natural units and varied (eg, kg, kWh, L, and CO_2). Thirteen of the 48 records measured environmental impacts using process-based life cycle analysis (LCA).[34,35,39,48,49,55,56,60-64,67] Process-based LCA is a method for estimating the impact of products over their entire life cycle, including from material extraction, manufacture, use, and disposal. Twelve records relied on previously published estimates of environmental impact.[26,33,37,38,41,42,51,52,54,65,68,70] Most of the remaining records used carbon-footprinting methods that fall short of process-based LCA, measuring 1 or few impacts from 1 stage of the life cycle of the intervention's products. One record converted cost into carbon footprint using the NHS carbon intensity estimate of 0.566 kg CO_2-e per GBP.[72]

Most records valued environmental impacts in carbon dioxide equivalent units (CO_2-e). Ten of the 48 records reported methods to value environmental impacts in monetary (n = 8) or health (n = 3) units.[33-35,40-42,52,55,56,59] This was mainly done by multiplying the environmental impact estimate by a published conversion factor. For monetary conversions, records used the social cost of carbon, non-traded price of carbon, or other published rates.[33,40-42,52,55,56,59] One record converted CO_2-e into disability-adjusted life years (DALY) using "damage factors" (this study also converted impacts into monetary units),[42] and another 2 estimated DALYs directly using life cycle impact analysis.[34,35] Further details are available in Table 1.

Full details of the environmental impacts identified and how they were measured and valued for each included record are provided in Appendix C in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.02.019.

Methods to Include Environmental Impacts in HTA or Economic Evaluations

Eleven records discussed or applied methods to include environmental impacts in an economic evaluation or HTA,[22,33-35,40,44,72-76] and these are described Appendix C in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.02.019. Each method is described in detail below and summarized in Table 2. Records that applied a method are further summarized in Table 3.

Information conduit

Information conduit methods were discussed by Toolan et al.,[22] who described a method in which environmental impacts are reported in recommendations or decisions but are not considered in the deliberation process and do not influence the decision made. We did not identify an application of this method.

Integrated evaluation

We identified several methods of performing an integrated evaluation based on cost-effectiveness (CEA), cost-utility (CUA), and cost-benefit analysis study designs, which differ depending on whether health outcomes are valued in monetary or health units. Five records described methods to fully incorporate environmental impacts into a CEA or CUA on the cost side of the equation in monetary units.[22,33,40,75,76] De Preux and Rizmie performed a CUA (cost per quality-adjusted life year) of different hemodialysis modalities and

included the carbon footprint of the alternatives as an additional monetary cost on the numerator of the incremental cost-effectiveness ratio (ICER).[33] Hensher adopted a similar approach in a CUA of oral antiglycemic medication.[40] Two records described a method to include environmental impacts on the outcomes side of the equation in health-related quality-of-life measures.[22,74] This was also referred to as an "enriched" analysis. Three records described methods to include environmental impacts in a cost-benefit analysis,[22,74,75] a type of economic evaluation where the benefits of the technologies being evaluated are converted into monetary units.[77] One record described a method whereby the explicit willingness-to pay-for health gains could be adjusted to reflect the environmental impacts of the technology being assessed.[74]

Parallel evaluation

We identified 5 methods of parallel evaluation: the calculation of the incremental carbon footprint effectiveness ratio (ICFER),[72] the calculation of the incremental carbon footprint cost ratio (ICFCR),[72] multi-criteria decision analysis (MCDA),[22,40,74,75] via existing HTA deliberation processes,[22,73,75,76] and alongside economic evaluations.[34,35,44]

Kindred et al.[72] proposed 2 new statistics: the ICFER and the ICFCR. The ICFER is the difference in carbon footprint between the intervention and its control divided by the difference in their effect (reported as CO_2-e per unit of the measure of effect). The ICFCR is the difference in in carbon footprint between the intervention and its control divided by the difference in their cost (reported as CO_2-e per dollar). The authors applied both these methods in a cost-utility analysis of obesity treatments.[72]

MCDA is a method for making decisions in which there are multiple competing criteria. Most MCDA are based on weighted-sum models, in which the relevant criteria (including environmental impacts and other factors, such as health, financial, and social impacts) are weighted based on their importance to the decision maker, and then each alternative is given a score for each criterion.[75] The preferred intervention is the one with the higher overall score. An MCDA that included environmental impacts was not identified in the reviewed records.

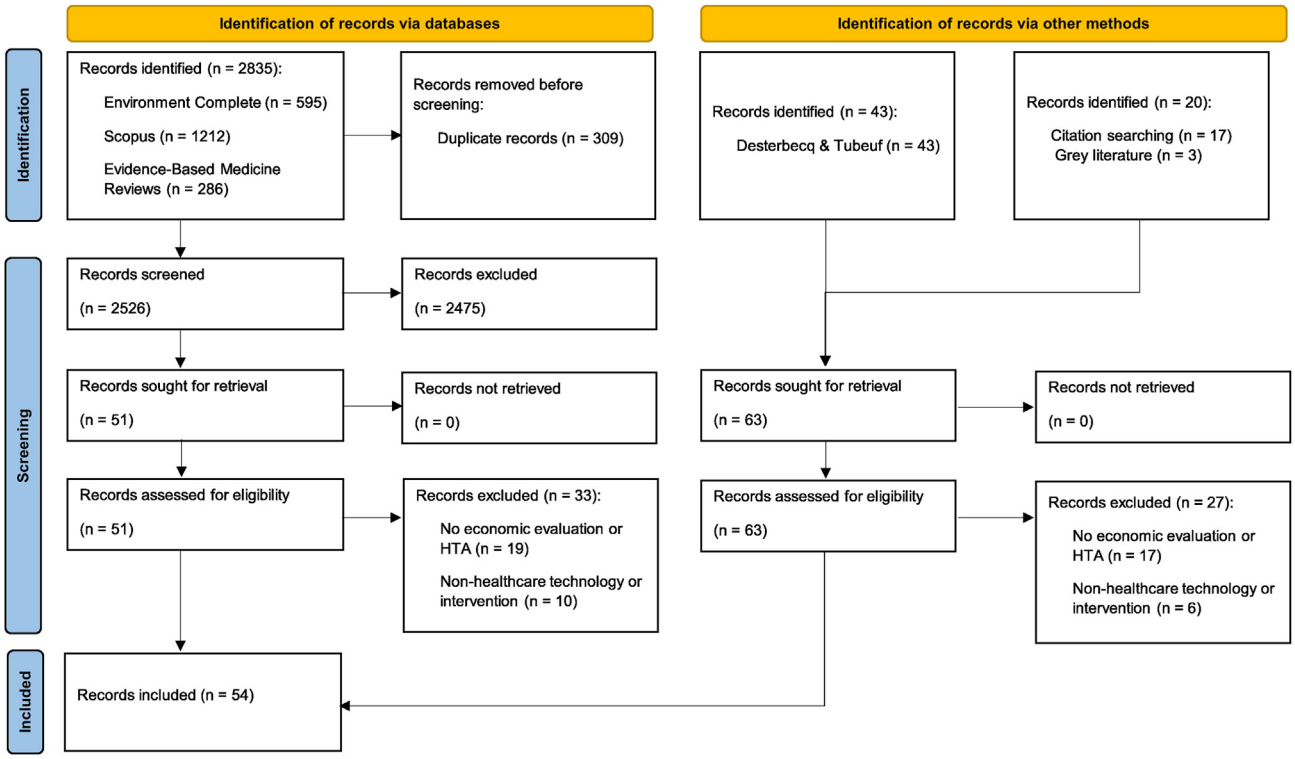
Methods to include impacts via existing HTA deliberation processes focused on allowing decision makers to freely consider relevant impacts when making assessments of technologies. Existing deliberation processes involve assessment of clinical effectiveness and cost-effectiveness of the new technology relative to its comparator, with discussion undertaken among expert committee members. Evidence about other outcomes such as environmental impacts may be used to modify such assessments.

Finally, Marsh et al.[44] used a Markov model to estimate quality-adjusted life years, costs, and environmental impacts of adding insulin to an oral antiglycemic treatment regime. Debaveye et al.[35] reported environmental impacts in DALYs as part of the "human health burden" of mebendazole for the treatment on soil-transmitted helminthiases in Vietnam and of antipsychotics for the treatment of schizophrenia in Belgium.[34] In all 3 examples environmental impacts were not included in an incremental cost-effectiveness ratio or other statistic.

Environment-focused evaluations

Environment-focused evaluations were described by Toolan et al.[22] but were not described or applied in any of the other reviewed records.[22] This kind of assessment would only consider the environmental impacts of the technology.

Figure 1. PRISMA Flow Diagram.



PRISMA indicates Preferred Reporting Items for Systematic Reviews and Meta Analyses.

Table 1. Valuing the environmental impacts of healthcare in monetary or health units.

Author	Year	Context	Value conversion method	Conversion
Debaveye et al[35]	2018	Soil-transmitted helminthiases	The public health burden associated with the intervention was measured in DALYs using LCIA.	No conversion used. DALYs estimated directly using LCIA.
Debaveye et al[34]	2019	Schizophrenia	The public health burden associated with the intervention was measured in DALYs using LCIA.	No conversion used. DALYs estimated directly using LCIA.
de Preux & Rizmie[33]	2018	Hemodialysis	Greenhouse gas emissions obtained from literature and converted from CO_2-e to GBP using non-traded price of carbon.	52 GBP per ton CO_2-e.
Hensher[40]	2020	NA	Provides examples of existing conversions from CO_2-e to monetary units using the social cost of carbon.	Conversions listed range from 0 GBP to 320 CAD.
Kponee-Shovein et al[42]	2022	Asthma inhalation therapy	Greenhouse gas emissions of inhalers obtained from previous published work and converted from CO_2-e to DALYs using "damage factors" estimated by Tang et al. Also converted to US Dollars using SCC published by United States Government Interagency Working group on Social Cost of Greenhouse Gases.	2.0×10^{-7} DALY per kg CO_2-e 6.2×10^{-7} DALY per kg CO_2-e Various SCC conversions as detailed by working group.
Kponee-Shovein et al[41]	2022	Asthma inhalation therapy	Greenhouse gas emissions of inhalers obtained from previous published work and converted from CO_2-e to GBP using non-traded price of carbon.	71 GBP per ton CO_2-e
Minotti et al[52]	2022	Food and diet	Greenhouse gas emissions of 2 diets was estimated and converted to Euros using the European Union Emissions Trading Scheme price on carbon.	0.82 EUR per kg CO_2-e
Ortsäter et al[56]	2019	Inhalation therapy for COPD and asthma.	Greenhouse gas emissions measured using life cycle analysis and converted from CO_2-e to US Dollars using the social cost of carbon.	50 USD per ton of CO_2-e.
Ortsäter et al[55]	2020	Inhalation therapy for COPD and asthma.	Greenhouse gas emissions measured using life cycle analysis and converted from CO_2-e to Euros using the social cost of carbon.	40 EUR per ton CO_2-e.
Rey Aldana et al[59]	2021	Telehealth for referring patients between primary care and hospital-based cardiology services.	Patient travel identified and measured. Carbon footprint of travel converted from CO_2 into Euros using conversion rate published by Litman.	0.0770235 EUR per gram of CO_2.

CAD indicates Canadian dollar; CO_2-e, carbon dioxide equivalent; COPD, chronic obstructive pulmonary disease; DALY, disability-adjusted life years; EUR, Euro; GBP, Pound sterling; GHG, greenhouse gases; LCIA, life cycle impact analysis; NA, not applicable; SCC, social cost of carbon; USD, US dollar.

Table 2. Methods to include environmental impacts in health economic evaluations and HTA.

Approach	Method	Description	Benefits and Limitations
Information conduit	Re-publish environmental impacts with HTA decision without further assessment.	Report environmental impacts during the HTA process but do not consider them as a reason for recommending/funding a technology. The information would be made available to the public.	Makes environmental impacts available to the public but unlikely to influence decision making.
Integrated evaluation	Include environmental impacts with monetary costs in an economic analysis.	Environmental impacts are converted to monetary units using published conversion rates and added to other costs in a CEA, CUA, or CBA. See de Preux & Rizmie, and Hensher for examples.	Internalizes environmental impacts (a negative externality) back into the evaluation. CEA, CUA, and CBA are well-established methods that are frequently used by HTA bodies globally to inform decisions. However, the methods may not lead to significant changes to the ICER, and results rely on accuracy of financial conversion rates used.
	Include environmental impacts with health outcomes in an "enriched" economic evaluation.	Environmental impacts are converted to health units and included on the outcomes side a CEA or CUA.	It is difficult to measure the marginal health changes associated with the environmental impacts of a particular intervention. This method also does not capture non-health impacts of environmental damage.
	Adjust willingness-to-pay thresholds	WTP modified to reflect environmental impact.	Some precedent with certain health conditions (eg, rare diseases and end of life). Method would not incentivize technologies with equal health impact but lower environmental impact.
Parallel evaluation	Calculate incremental carbon footprint effectiveness ratio (ICFER).	Environmental impacts are used to calculate an ICFER, which is the difference in environmental impact divided by the difference in effectiveness. See Kindred et al for an example.	Can be done in parallel with economic evaluation without conversion of environmental impacts into monetary or health units. Interpretation of statistic difficult in scenarios where a trade-off between health outcomes and environmental impacts takes place.
	Calculate incremental carbon footprint cost ratio (ICFCR).	Environmental impacts are used to calculate an ICFCR, which is the difference in environmental impact divided by the difference in cost. See Kindred et al for an example.	Can be done in parallel with economic evaluation without conversion of environmental impacts into monetary or health units. Monetary thresholds might be available based on existing carbon prices.
	Consider environmental impacts in HTA deliberative decision making process.	Consider environmental impacts freely during the HTA deliberation process.	Allows decision makers to choose importance of impacts on a case-by-case basis. Method places an onus on those undertaking the decision to understand environmental impacts and their significance. Environmental experts should therefore be included in HTA panels that use this method.
	Multi-criteria decision analysis (MCDA).	Include environmental impacts as one criterion of a MCDA alongside other factors important to decision makers.	Application of MCDA to HTA is in its infancy and no identified examples in the HTA context that include environmental impacts were identified. Places the burden of understanding the value of environmental impact changes on the decision makers involved in the MCDA (which may lead to proxy bias).
	Publish environmental impacts with economic evaluation without further assessment.	Estimate and report environmental impacts during economic evaluation but do not include them with other cost or health information or in a final calculated ICER. This could be done in a cost-consequence analysis. See Marsh et al. and Debaveye et al. for examples using an economic model.	Does not require changes to economic evaluation methods and allows decision makers to see environmental information separate to economic information. May not influence decision making.

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Table 2. Continued

Approach	Method	Description	Benefits and Limitations
Environment-focused evaluations	Consider interventions with environmental benefits but no health or cost benefits.	Require HTA agencies to consider only environmental impacts in their deliberations for interventions with no health or cost benefits.	Encourage the development of low-impact technologies with no health or cost benefits. Environmental experts would need to be included in HTA panels to properly consider the submitted information.

Note. Methods categorized by approach as proposed by Toolan et al.[22]
CBA, cost-benefit analysis; CEA, cost-effectiveness analysis; CUA, cost-utility analysis; HTA, health technology assessment; ICER, incremental cost-effectiveness ratio; MCDA, multi-criteria decision analysis; WTP, willingness-to-pay.

Discussion

We identified 10 methods to include environmental impacts in health economic evaluations and HTA. One is an information conduit approach, in which environmental impacts are published with HTA decisions without further assessment. Three are integrated evaluations, in which environmental impacts are fully incorporated into existing economic evaluations as either a cost or outcome, or by adjusting the willingness-to-pay threshold. Five are parallel evaluations, in which environmental impacts are included in an ICFER or ICFCR ratio, multi-criteria decision analysis, deliberative HTA decision making process, or alongside an economic evaluation. One is an environment-focused approach.

The overall approach for considering environmental impacts of a health technology follows 4 steps: identify, measure, value, and include in HTA. The first step is to identify relevant impacts (Fig. 2). Although healthcare causes diverse environmental impacts, most work has focused on greenhouse gas emissions, particularly from energy use and travel. This is likely because of the relative ease of calculating emissions from these activities. McAllister et al.[75] argue that estimating greenhouse gas emissions should be the priority of current research efforts in order to limit human induced climate change. We agree with this prioritization and note that identified methods also allow for the consideration of other environmental impacts if this is desired. A recent review by Desterbecq and Tubeuf summarizes different environmental dimensions of healthcare that have been considered in economic evaluations, focusing on carbon dioxide, water, energy, and waste.[78]

The second step in evaluating environmental impacts is to measure them in natural units or obtain an estimate of their impact from existing sources (Fig. 2). Process-based LCA is a method for estimating the environmental impact of products over their entire lifecycle and has emerged as the preferred method for measuring the impacts of healthcare technologies. Making LCA estimates of clinical pathways is time consuming, resource intensive, and requires technical expertise. It is therefore unlikely to be feasible to estimate this alongside every economic evaluation. Instead, environmental scientists may establish LCA estimates for common healthcare procedures and pathways that can be used by health economists in their evaluations (see, for example, estimates of pathology testing[79] and diagnostic imaging[80] in Australia). McAlister et al.[75] recommends that databases of these LCA estimates should be developed to facilitate future research, and this should be country and time specific (see, for example, HealthcareLCA[81]). Clinical trialists should also ensure they are collecting sufficient information to allow these estimates to be calculated and may consider collaborating with environmental scientists to undertake process-based LCAs alongside measurement of other outcomes as necessary.

The third step in evaluating environmental impacts is to value them (Fig. 2). The preferred method for valuing environmental impacts was in CO_2-e units, which is the global standard reference unit for greenhouse gases (GHG) emissions, as well as a common output unit of LCA. Estimating environmental impacts in CO_2-e is beneficial because it allows multiple different impacts, each with their own natural units of measurement and global warming potential, to be converted into a single figure that can be reported to HTA bodies. To include impacts in an integrated evaluation, environmental impacts may be converted into monetary units or into DALYs. In some jurisdictions, such as the United Kingdom[82] and United States,[83] monetary valuations of GHG emissions are estimated by the government for use in policy decisions but others, such as Australia, do not. Research in these contexts will need to make use of relevant options from the literature. Care must be taken to ensure accurate and useful conversion rates are used. DALY conversions are less established than monetary conversions, and further work is needed to understand their application. DALY use is potentially limited by only capturing the health-related impacts of GHG emissions and not the broader societal impacts of increasing temperatures.

Once environmental impacts have been identified, measured, and valued, they can be included in an HTA or economic evaluation using 1 of the 10 methods identified in this review. If environmental impacts are included as an information conduit then 1 method is available to do so: reporting environmental data alongside HTA decisions. Because the environmental impacts are not considered in the HTA deliberation, an information conduit approach is unlikely to influence funding decisions. But by providing public information on the environmental impacts of healthcare, this approach may support clinicians and patients to choose low-carbon care.

If environmental impacts are to be included in an integrated evaluation, the most researched option is to convert them into monetary units and include them in the cost side of a cost-effectiveness or cost-utility analysis. This method was applied by 2 records and appears to be feasible, so long as appropriate monetary conversions are identified. This method is beneficial because it internalizes the negative externality of environmental impacts back into the economic decision. This method could also complement the inclusion of other societal costs into economic evaluations (such as productivity losses) by taking a societal perspective. In this way a more complete estimate of the full economic cost of healthcare interventions is provided. Further applications of this method in diverse contexts are needed to understand how ICER estimates might change after the inclusion of environmental impacts and what (if any) changes to cost-effectiveness thresholds might be needed.

Alternatively, environmental impacts can be included in an integrated evaluation by converting them to health units and

Table 3. Studies demonstrating methods to include environmental impacts in health economic evaluations and HTA.

Author	Year	Intervention and Comparator	Method	Approach
de Preux and Rizmie[33]	2018	Home hemodialysis vs in-center hemodialysis	Cost-utility analysis with environmental impacts converted from CO_2-e to dollars and included with other costs.	Integrated evaluation
Debaveye et al[35]	2018	Anthelmintic treatment of hookworm with mebendazole vs no treatment	DALYs associated with the pharmaceutical supply chain were measured using LCIA.	Parallel evaluation
Debaveye et al[34]	2019	Pharmaceutical treatment of schizophrenia with paliperidone palmitate vs treatment interruption	DALYs associated with the pharmaceutical supply chain were measured in using LCIA.	Parallel evaluation
Marsh et al[44]	2016	Adding insulin to an oral antiglycemic medication treatment regime vs oral antiglycemic medication alone	Environmental impacts included in a cost-utility analysis model (IMS CORE diabetes model). Model estimated impacts but did not include them in an ICER or further evaluation.	Parallel evaluation
Hensher[40]	2020	Adding insulin to an oral antiglycemic medication treatment regime vs oral antiglycemic medication alone	Used work by Marsh et al. to calculate an ICER.	Integrated evaluation
Kindred[72]	2024	Semaglutide vs bariatric surgery for the treatment of obesity.	Cost-utility analysis using a Markov model. Environmental impacts used to calculate the ICFCR and ICFER alongside the ICER.	Parallel evaluation

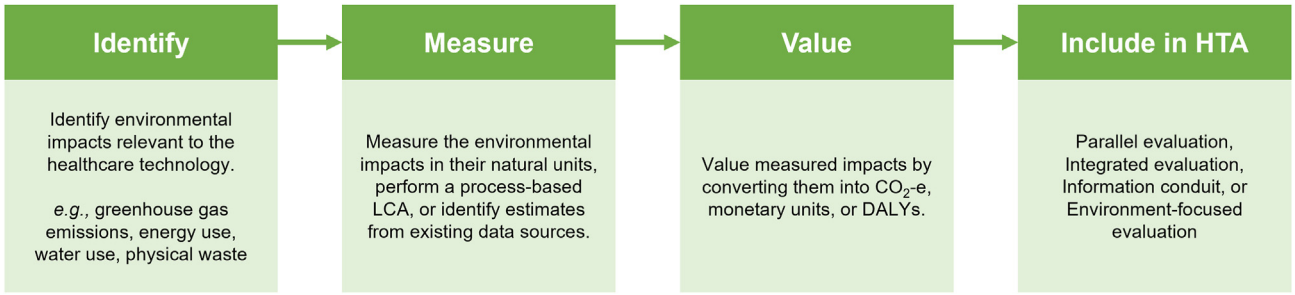
CO_2-e, carbon dioxide equivalent; HTA, health technology assessments; ICER, incremental cost-effectiveness ratio; ICFCR, incremental carbon footprint cost ratio; ICFER, incremental carbon footprint effectiveness ratio; LCIA, life cycle impact analysis.

including them on the outcomes side of the economic evaluation. As noted above, information about including environmental impacts as a health outcome, instead of a cost, in an economic evaluation is limited. We did not identify any applications of how to do this in a CEA or CUA. Although Kponee-Shovein et al.[42] converted environmental impacts into DALYs and Debaveye et al.[35] estimated DALYs using life cycle impact analysis,[34] neither were fully integrated into an evaluation.[84] Further research is needed to demonstrate and evaluate these methods if they are to be widely used. To avoid double counting, environmental impacts should be included on either the cost or outcome side of an economic evaluation, but not both.

The final integrated evaluation method identified was to adjust explicit willingness-to-pay thresholds, but few details were given for how this could be done in practice. A similar approach is used in end-of-life care in the United Kingdom,[85] and this may inform how the approach could work for environmental impacts. Future research might identify the amount decision makers are willing to pay for clinically equivalent healthcare with lower environmental impacts.

If researchers do not wish to integrate environmental impacts directly into the economic evaluation, then a parallel evaluation approach might be preferred. The simplest parallel evaluation methods were proposed by Kindred et al.[72] and involve reporting 2 new statistics alongside the ICER in an economic evaluation: the ICFER and ICFCR. The ICFER and ICFCR attempt to summarize the environmental impact of the proposed health technology without changes to existing health economic methods. The ICFER summarizes the environmental "cost" of health gains, with a positive ICFER indicating that health gains are associated with increased environmental impacts. The authors highlight a controversial scenario in which both environmental impact and health

Figure 2. Methods to include environmental impacts in HTA.



HTA indicates health technology assessment.

outcomes might decrease, raising the question of whether such a trade-off would ever be considered by policymakers. The ICFCR is a similar statistic that indicates whether environmental impacts increase or decrease per dollar spent. There are currently no established thresholds for either of these statistics, and further work is needed in this area if they are to be widely used.

Another parallel evaluation approach identified in this review was to include environmental impacts in an MCDA. MCDA has been used to support HTA decisions,[86] but no examples considered environmental impacts, instead focusing on various other health, economic, and social implications. Further research is needed to investigate MCDA methods as a way of including environmental impacts in HTA, including work to establish the weight applied to environmental impacts.

Another parallel evaluation identified was to consider impacts in the HTA deliberation process. This is an appropriate approach in jurisdictions that allow environmental impacts to be considered, but it is difficult to know exactly how this will be done and whether decision makers are capable of interpreting environmental impact information to make such decisions. Finally, environmental impacts can be reported alongside economic evaluations but separate to other cost and health outcomes. This could be done using a cost-consequence framework in which all costs and effects are reported separately.

The methods described above may be prone to "green washing," in which the impacts of the technology might be minimized or the reduction in impacts exaggerated to make it more likely that it receives a favorable HTA outcome. Future research should consider this problem and how it might be addressed. Although research in this area has significantly increased in recent years, more work is needed before any method can be widely used by health economists or HTA agencies. Other scoping reviews on this topic have also identified the need for further methodological work to overcome challenges in the use of available methods.[87,88] The review by Pinho-Gomes et al.[87] identifies frameworks and methods for including environmental considerations in guidelines and HTAs. The review by Guiardo-Fuentes et al.[88] identifies challenges of incorporating environmental impacts into economic evaluations and HTAs.

This field of research is rapidly evolving, and our results are up-to-date to the stated search period. Several records were identified via hand searching, and this may be due to a limitation in our search strategy. However, we are confident we identified all relevant methods described so far.

Conclusion

We identified 10 methods to include environmental impacts in economic evaluations and HTA based on information conduit, integrated evaluation, parallel evaluation, and environment-focused evaluation approaches. No method is currently fit for widespread use, and further work is needed to determine which of these methods is best suited to aid decision makers for choosing low-impact healthcare interventions.

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Author disclosure forms can be accessed below in the Supplemental Material section.

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